

Maxell G. Milay

AI & Machine Learning Engineer

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SUMMARY

AI & Machine Learning Engineer with proven expertise in architecting, deploying, and maintaining production-grade AI systems from concept to continuous operation. Specialized in designing enterprise-scale LLM orchestration frameworks, multi-agent workflows, and RAG-powered applications using OpenAI models, LangChain, and AWS cloud infrastructure. Demonstrated track record in building complete ML pipelines encompassing data ingestion, model development, CI/CD/CT automation, deployment, and observability—reducing operational costs while maintaining high system availability. Experienced in translating complex business requirements into scalable technical solutions, implementing cost-effective prompt engineering strategies, and establishing monitoring dashboards for real-time performance tracking. Proficient in MLOps best practices with specialized knowledge in Google Cloud Platform (GCP) services including Vertex AI, Cloud Run, and ML pipeline orchestration. Passionate about automating the full ML lifecycle to ensure seamless transitions from experimentation to production deployment.

EXPERIENCE

ECLARO

Quezon City, Metro Manila

AI Engineer

April 2025 - Present

- Architected enterprise-scale Generative AI orchestration framework leveraging OpenAI GPT-4/GPT-4o models with AWS Lambda and Step Functions, enabling secure LLM deployment across 5+ Insurance Submission business units processing 10,000+ monthly documents
- Pioneered multi-agent workflow prototypes using LangChain and LangGraph frameworks, demonstrating 40% reduction in processing time for underwriting workflows and 35% improvement in decision consistency across test scenarios
- Championed Model Context Protocol (MCP) adoption by developing 6+ custom connectors integrating internal policy APIs, claims databases, and document repositories, enabling contextual AI assistants with secure enterprise data access
- Implemented comprehensive AI observability stack using AWS CloudWatch, CloudWatch Logs Insights, and custom metrics dashboards, achieving 99.5% framework uptime and reducing incident detection time by 60% through real-time anomaly alerts
- Redesigned Intelligent Document Processing (IDP) pipeline using Amazon Textract and custom NER models, improving field-level extraction accuracy from 78% to 92% for complex insurance forms and reducing manual review requirements by 45%
- Engineered AI-driven information extraction system for First Notice of Loss (FNOL) claims processing using spaCy NLP pipelines and transformer-based entity recognition, achieving 88% accuracy across 15+ structured field types
- Translated complex insurance domain requirements into technical specifications for 8+ ML models, collaborating with product owners and underwriters to define extraction schemas, validation rules, and confidence thresholds
- Optimized LLM inference costs through prompt engineering, response caching, and model selection strategies (GPT-4 vs GPT-4o-mini), reducing API expenses by 30% while maintaining 90%+ response quality benchmarks

Insight Genie

Singapore, Singapore

Data Scientist Intern

June 2025 - September 2025

- Engineered alternative credit scoring features using digital footprint analysis, leveraging eKYC verification and reverse lookup capabilities to expand financial inclusion coverage by 40%
- Researched and integrated open-source Sentiment Emotion Recognition (SER) models, improving sentiment classification accuracy by 35% through systematic model evaluation and hyperparameter optimization
- Developed and validated voice analysis ML models for production deployment, testing 12+ model architectures to ensure reliability and reproducibility across diverse call center datasets
- Streamlined ML experimentation workflows using MLflow framework, reducing model training iteration time by 45% and enabling systematic hyperparameter tracking across 50+ experiments
- Designed and deployed a Streamlit-based voice model creation interface, simplifying the LLM API integration process and reducing non-technical user setup time from 2 hours to 15 minutes

BitWork Solutions

Florida, Miami

Lead Software Engineer

December 2024 - April 2025

- Led cross-functional team of 5 engineers in architecting and delivering an image processing mobile application from concept to production, achieving 95% sprint velocity across 8+ development cycles
- Orchestrated agile sprint planning and backlog prioritization for 50+ user stories, maintaining 90% on-time delivery rate and reducing scope creep by 40% through structured requirements gathering
- Spearheaded critical backend migration from Directus headless CMS and AWS Lambda to scalable Django REST API architecture with PostgreSQL, reducing infrastructure costs by 35% and improving API response times by 50%
- Established code quality standards through comprehensive pull request reviews and enforced best practices, reducing production bugs by 65% and technical debt accumulation by 55%
- Implemented CI/CD pipelines with automated testing and deployment workflows, decreasing release cycle time from 2 weeks to 3 days

Part-time DevOps Engineer

July 2024 - December 2024

- Architected and deployed CI/CD pipelines with automated linting and testing across 3 enterprise projects, reducing manual QA time by 70% and catching 85% of bugs pre-production
- Engineered AWS cloud infrastructure integrating S3 and DynamoDB for scalable image data storage and retrieval, which was projected to support 50,000+ daily image uploads with 99.9% uptime
- Re-architected legacy backend to serverless AWS Lambda functions, achieving 60% reduction in server costs and enabling automatic scaling from 100 to 5,000 requests per minute
- Implemented infrastructure-as-code practices using Terraform, reducing environment provisioning time from 4 hours to 15 minutes

BPOSeats

Mandaue City, Cebu

AI Engineer

June 2024 - January 2025

- Architected and deployed machine learning-powered productivity forecasting system processing 15,000+ historical task records, projected to increase operational efficiency by 55% and streamline resource allocation workflows
- Engineered advanced prompt engineering strategies for OpenAI GPT models, optimizing API token consumption and reducing LLM operational costs by 40% through context window optimization and response caching techniques
- Co-led innovative AI-driven effort estimation framework integrating NSGA-II genetic algorithms with LLM reasoning, improving task duration prediction accuracy by 30% over baseline estimation methods
- Conducted comprehensive research in software effort estimation methodologies spanning genetic algorithms, transformer models, and Monte Carlo simulations, presenting findings to technical leadership team
- Implemented enterprise-wide LLM cost monitoring dashboard tracking token usage across 8+ features and API endpoints, identifying optimization opportunities that reduced operational expenses by 35%
- Designed real-time analytics visualizations for AI spending patterns using Python and Plotly, enabling data-driven decision-making for AI feature development and prioritization

Full Stack Web Developer

September 2023 - May 2024

- Pioneered company's first production-grade vector database implementation using PostgreSQL with pgvector extension, enabling semantic search capabilities across 5,000+ documentation entries with 85% retrieval accuracy
- Architected and deployed comprehensive testing infrastructure integrating Vue.js and Django unit tests into CI/CD pipelines, reducing production deployment failures by 80% and improving release confidence
- Developed intelligent RAG-powered customer support chatbot leveraging OpenAI embeddings and semantic search, projected to deflect 75% of tier-1 support tickets and reduce average resolution time by 40%
- Engineered automated ETL pipeline migrating 2,500+ legacy documentation pages to vector database, implementing chunking strategies and embedding generation that achieved sub-3-second response times
- Conducted systematic debugging and code optimization initiatives reducing critical bug count by 60%, contributing to 65% improvement in system availability (from 94% to 99%+ uptime)
- Implemented semantic similarity search algorithms enabling context-aware documentation retrieval with cosine similarity scoring and relevance ranking

Symph

Cebu City, Cebu

Web Developer Intern

December 2022 - March 2023

- Assisted in deploying code changes to 3+ client websites in collaboration with backend and DevOps teams, learning production deployment workflows and staging environment best practices
- Diagnosed and resolved 15+ UI/UX issues across 2 active client projects, improving page load performance by 25% and enhancing mobile responsiveness for 5+ key user flows
- Implemented client-requested feature enhancements and bug fixes using HTML, CSS, JavaScript, and React, contributing to 20% increase in client satisfaction scores
- Collaborated with cross-functional teams in agile sprint cycles, participating in code reviews and daily standups to ensure delivery quality and adherence to project timelines

VOLUNTARY WORK EXPERIENCE

AI Pilipinas Cebu Chapter

Cebu City, Cebu

AI Infrastructure and Deployment Advocate

July 2024 - Present

- Researched relevant techniques in MLOps, including ML workflows, from [data ingestion and preprocessing](#), [feature engineering](#), [model development and training](#), model evaluation, deployment to production, and performance monitoring
- Created [Medium articles](#) revolving around Google Cloud's AI tools, including Vertex AI and other cloud services for MLOps and ML workflows

University of the Philippines Computer Science Guild

Cebu City, Cebu

Lead Software Engineer

November 2023 - Present

- Led a team of 7 designers and 10 developers in creating multiple versions of the official UP Computer Science Guild Website
- Established the entire workflow between designers and developers, rooting from relevant agile principles
- Deployed the website with CI/CD and GitHub Actions through Docker on university servers
- Managed and reviewed the UX of designers and code of each developer, ensuring code quality and performance

LEADERSHIP, PROJECTS, & RECOGNITION

Beacon

- Created a career guidance application that utilizes generative AI to provide users with personalized visual roadmaps towards their viable career option based on their user profile information.
- Led a team of 5 in building a working prototype of the application within the span of 6 hours
- Implemented a storage system and authentication through Google Firebase and leveraged OpenAI LLM models
- Garnered the 1st Place of the first UP Computer Science Guild Komsai Week Hackathon in 2024

Finite Automaton Visualizer

- Created a web application that generates DFA graphs based on regular expressions, and also verifies string inclusion by visually simulating the algorithm through a user-friendly and interactive interface.
- Led a team of 4 in creating a working prototype of our application through agile development
- Implemented the core algorithm which involved generating an abstract syntax tree from a regular expression in consideration of precedence

Lakbai

- Created a GIS application that uses a dynamic, data-driven pedestrian accessibility index powered by a fuzzy logic system for promoting sustainable and equitable urban mobility in the Philippines within the span of 2 weeks
- Awarded as the champion of the 2024 Philippine Junior Data Science Challenge hosted by the UP Data Science Society
- Led the integration of the fuzzy inference system and multilayer perceptron into the application
- Deployed the frontend using Vercel, the backend using Google Cloud Run, and the postgresql database using Google Cloud SQL
- Setup CI/CD pipelines using Github Actions that tied together GCP services for deployment and integration

EDUCATION

University of the Philippines Cebu

Cebu City, Cebu

Bachelor of Science in Computer Science

- Led 3 teams through small and large scale competitions and projects, achieving the best competition places and highest ratings

Philippine Science High School - Central Visayas Campus

Argao, Cebu

STEM Strand

High Honors & Excellency in Physics

- **Project DALOY** - Drone-based Depth and Atmospheric Level Open-Source monitoring device for the Filipino Youth
- **Thesis** - Simulated Biomimicry of Photovoltaic Tree Architecture Based on the Phyllotaxy and General Tree Crown Shape of *Pinus Strobus* (Eastern White Pine)

CERTIFICATIONS

Machine Learning Specialization

DeepLearning.AI & Stanford

- Completed 3 hands-on courses involving supervised, unsupervised, and reinforcement learning, along with neural networks using TensorFlow and Scikit-learn frameworks
- Replicated the [Lunar lander reinforcement](#) learning problem and solved through Deep Q Learning using the Proximal Policy Optimization (PPO)
- Created a [movie recommender system](#) implemented using vectorization and similarity search of metadata
- Built an unsupervised [customer segmentation model](#) using K-means clustering using each customer's spending score and yearly income
- Built and compared different [prediction models for house pricing](#) such as Linear, Random Forest, XGBoost, CatBoost regressors, and more

Deep Learning Specialization

DeepLearning.AI & Stanford

- Completed 5 hands-on courses around neural networks, and learned various optimization techniques, along with ML development iteration best practices
- Built industry grade CNNs and RNNs from scratch like [VGG-16/19](#) and variations of the [MobileNet](#), achieving 77% and 78% validation accuracies respectively both using the CIFAR-10 dataset